1. Input data

- Design moment M_{Ed} kNm
- Section: b mm, d mm, d' mm
- Materials: f_{ck} MPa, f_{cd} MPa,f_{yk} MPa, f_{yd} MPa
- α = 1.0 (Estonian NA)

2. Step-by-step algorithm

Step 0 – Preliminary check

If $M_{Ed}=0 \Rightarrow$ provide only minimum reinforcement (go to Step 6).

Step 1 – Calculate relative moment μ

$$\mu = \frac{M_{Ed}}{\alpha f_{cd} b d^2}$$

Step 2 – Determine $\lambda, \eta, \xi_{\lim}, \mu_{\lim}$

If $f_{ck} \le 50$: λ=0.8 , η=1.0, $\varepsilon_{cu3}=0.0035$

Else ($50 < f_{ck} \le 90$): $\lambda = 0.8 - \frac{f_{ck}-50}{400}, \eta = 1 - \frac{f_{ck}-50}{200}, \varepsilon_{cu3} = 0.0026 + 0.035\left(\frac{90-f_{ck}}{100}\right)^4$

$\xi_{\lim} = \frac{\varepsilon_{cu3}}{\varepsilon_{cu3}+\varepsilon_{yd}}$ $\xi_{\lim 2} = \frac{\varepsilon_{cu3}}{\varepsilon_{cu3}-\varepsilon_{ycd}}$ $\varepsilon_{yd} = \varepsilon_{ycd} = \frac{f_{yd}}{E_s}$, where $E_s = 200\,000\,MPa$

$\omega_{\lim} = \eta \lambda \xi_{\lim} \Rightarrow \mu_{\lim} = \omega_{\lim}\left(1 - \frac{\omega_{\lim}}{2}\right)$

Step 3 – Check against ductility limit $\mu_{\lim}$

- If $\mu \le \mu_{\lim} \Rightarrow$ singly reinforced section ⇒ go to Step 4.
- If $\mu > \mu_{\lim} \Rightarrow$ compression reinforcement needed (doubly reinforced) ⇒ go to Step 5.

Step 4 – Singly reinforced section ($\mu \le \mu_{\lim}$)

Calculate mechanical reinforcement ratio: $\omega = 1 - \sqrt{1-2\mu}$

Required tension steel area:

$$A_{s,req} = \frac{\alpha \omega f_{cd} b d}{f_{yd}}$$

Step 5 – Doubly reinforced section ($\mu > \mu_{\lim}$)

1. Calculate the "balanced" moment the concrete alone can carry: $M_{Rd,\lim} = \mu_{\lim}\, \alpha\, f_{cd}\, b\, d^2$
2. Excess moment that must be carried by the steel couple: $M_{Ed}' = M_{Ed} - M_{Rd,\lim}$
3. Compression reinforcement (top steel): $A_s' = \frac{M_{Ed}'}{f_{ycd}(d - d')}$
4. Tension reinforcement (bottom steel) area: $A_s = A_{s,\lim} + A_s'$ where $A_{s,\lim}$ is the tension steel that corresponds to $\mu_{\lim}$.

Step 6 – Minimum reinforcement check (EC2 9.2.1.1) – applied after Steps 4 or 5

If $f_{ck} \le 50$: $f_{ctm} = 0.3\, f_{ck}^{\frac{2}{3}}$

Else ($50 < f_{ck} \le 90$): $f_{ctm} = 2.12 \ln\left(1 + \frac{f_{ck} + 8}{10}\right)$

$$A_{s,min} = max\left(0.26 \frac{f_{ctm}}{f_{yk}} b_t\, d\,;\, 0.0013\, b_t\, d\right)$$

- b_t = width of tension zone (b or b_{eff})
- This value is **required on the tension face**.
- You must also provide minimum reinforcement on the **opposite face** (compression face) (typically 0.0013bd or at least 2 bars).

Step 7 – Final provided reinforcement

$$A_{s,prov} = max\left(A_{s,req}\,;\, A_{s,min}\right)$$

Step 8 – Verify capacity

Calculate the actual resisting moment of the provided reinforcement: $M_{Rd,prov} \ge M_{Ed}$

The check is done with the actual bars you have chosen.

$$x = \frac{f_{yd} A_s - f_{ycd} A_s'}{\eta f_{cd} b \lambda}$$

If $x < \xi_{c2} d'$: concrete alone resists compression ⇒ no compression steel needed.

$$x = \frac{f_{yd} A_s}{\eta f_{cd} b \lambda} \qquad M_{Rd} = \eta f_{cd} b \lambda x (d - 0.5 \lambda x) = f_{yd} A_s (d - 0.5 \lambda x)$$

If $x \ge \xi_{c2} d'$: concrete compression is insufficient ⇒ need compression steel.

$$M_{Rd} = \eta f_{cd} b \lambda x (d - 0.5 \lambda x) + f_{ycd} A_s' (d - d')$$

Finally, check if section is not over-reinforced $x \le \xi_{\lim} d$

If otherwise $M_{Rd} = \mu_{\lim}\, \eta f_{cd} b d^2 + f_{ycd} A_s' (d - d')$